ISYS2120 Assignment 3 (Concepts) (sem1, 2025)

Due: Sunday 2 November, 11:59pm Sydney time
Value: 5%
This assignment is **individual work**.

Late work policy: (from unit outline) Late submissions for assignments will incur a penalty of 5% of the maximum awardable marks for each day, or part-day, past the due date, up to a maximum of 7 days (as after this time, feedback on on-time submissions will be available, resulting in an unfair advantage if submissions after this time were accepted). After 7 days late submissions will not be accepted. Where special consideration is granted for these assessments, extensions of a maximum of 7 days will be permitted. After 7 days, reweighting of other relevant tasks will be applied.

Summary (but details further down are authoritative)
Done individually. You are allowed to use GenerativeAI, but if so, you must submit an appendix showing each prompt you used, and the corresponding output from the computer; you will be marked only for what you have contributed, not on what was done by the computer.
Provided for you: four questions, and a template for the cover page
You produce and submit [on Canvas]: a pdf document, with cover page as shown in the template, and then your answers to the four questions.
Marks: these will be released on Canvas, initially for students who submit at the original due date, and perhaps later for students whose submission is delayed eg for simple extension.

The questions

The questions A and C are all based on the following relational logical schema about music tracks.

```
CREATE TABLE Artist (
    AId INTEGER,
    AName VARCHAR(100) NOT NULL,
    ACountry VARCHAR(50),
    PRIMARY KEY (AId)
);
CREATE TABLE Song (
    SId INTEGER,
    STitle VARCHAR(100) NOT NULL,
    SDuration INTEGER, -- Seconds
    AId INTEGER,
    PRIMARY KEY (SId),
    FOREIGN KEY (AId) REFERENCES Artist(AId)
);
CREATE TABLE Playlist (
    PId INTEGER,
    PName VARCHAR(100) NOT NULL,
    PUser VARCHAR(30), -- created by
    PRIMARY KEY (PId)
);
CREATE TABLE PlaylistSong (
    PId INTEGER,
    SId INTEGER,
    AddedDate DATE,
    PRIMARY KEY (PId, SId),
    FOREIGN KEY (PId) REFERENCES Playlist(PId),
    FOREIGN KEY (SId) REFERENCES Song(SId)
);
CREATE TABLE GenreInfo (
    SId INTEGER,
    Genre VARCHAR(30),
    PopularityScore INTEGER,
    PRIMARY KEY (SId, Genre),
    FOREIGN KEY (SId) REFERENCES Song(SId)
);
```

In **Artist,** a row (10, ‘Snoop Dogg', 'US') represents an artist whose AId is 10, name is “Snoop Dogg” and the country is "US". In **Song**, a row (17, 'Gold Rush', 300, 10) represents that the song with SId 17 is “Gold Rush" that is 5 minutes long (60*5 seconds) performed by artist with AId 10. In **Playlist**, a row (11, 'Doggs', 'Richard') represents a playlist with 11 as playlist ID created by a user Richard. In **PlaylistSong**, a row (11, 17, '2023-11-02') represents that the song with SId 17 was added to playlist with PId 11 on '2023-11-02'. In **GenreInfo**, a row (17, 'funk', 11) represents that the song 17 has a popularity score of 11 in the genre "funk".

A [relational algebra] (1 point)
Note: if you have difficulty producing relational algebra symbols, you may use text, with SEL_{C} to represent σ_C and PROJ_{D} to represent π_D and X JOIN Y to represent X ⋈ Y.

Write *two different (but logically equivalent) relational algebra expressions* involving the tables in the schema above, so that each has a result which will answer the request ""Find the titles of all songs that appear in a playlist which was created by user Richard, and have duration less than 180 seconds"

B [relational design theory] (2 points)
Consider the following relational design, which collects data about sport events. There is a single table:
SportsInfo(Player,Team, Mascot, GDate, Score, Location, Fee).
In **SportsInfo**, a tuple 'Sally Snape', 'Sydney', 'swan', '30-03-2024', 56, 'Olympic Park', 15000) represents that Sally Snape played for the Sydney team whose mascot is a swan, in a game on 30 March 2024 held at Olympic Park , a game in which the team scores 56 and for which Sally was paid a fee of $15000.
The following functional dependencies are valid in this schema:

- **Player → Team**
- **Team → Mascot**
- **Team, GDate → Score, Location**
- **Player, GDate → Fee**

B(i) Based on the **SportsInfo** schema and functional dependencies given on page 13, explain in English the meaning of the functional dependency Team, GDate → Score, Location. Also, give an example of data that would not be allowed in the table **SportsInfo** because of this dependency.

B(ii) Based on this schema and its fds, calculate the attribute closure (Team, GDate)+ Show the step-by-step working of the calculation.

B(iii) State whether the relation **SportsInfo** is in BCNF. Justify your answer.

B(iv) Give a lossless-join decomposition of **SportsInfo** into two relations. For each decomposed relation, state the functional dependencies that hold for it, and state a primary key for it. Justify that your decomposition has lossless-join property.

C. [index choices] (1 point)
Consider again the relational schema shown above, and the GenreInfo table. Suppose that no CREATE INDEX commands have been executed; that means that the structure of the table is as a tree-based clustering index on the primary key (SId, Genre).

Consider the SQL query below.

```
SELECT Genre
FROM GenreInfo
WHERE PopularityScore < 4 AND SId = 17
```

C(i) State the steps through which the query would be answered with the initial structure of the table.

C(ii) Write SQL to create an index that will allow much faster execution for the query above, when the database has a large amount of data in the table. Also, state how the index would allow the query to be calculated.

D. [Database-backed web-app security] (1 point)

D(i) In the code we provided in lab for week 8, a security goal is that an end-user who does not have an isAdmin role, should not be able to modify information about end-users. Describe the mechanism(s) by which the system seeks to meet this goal, and discuss any vulnerability in the system regarding this security goal.
D(ii) Suppose you work for a company which has built a web application, based on the code we provided in lab for week 8. Write a description, for a non-technical manager to read, about what should be done to protect the information kept in the config.ini file, and what the consequences could be if the contents of file is ever leaked outside the company.